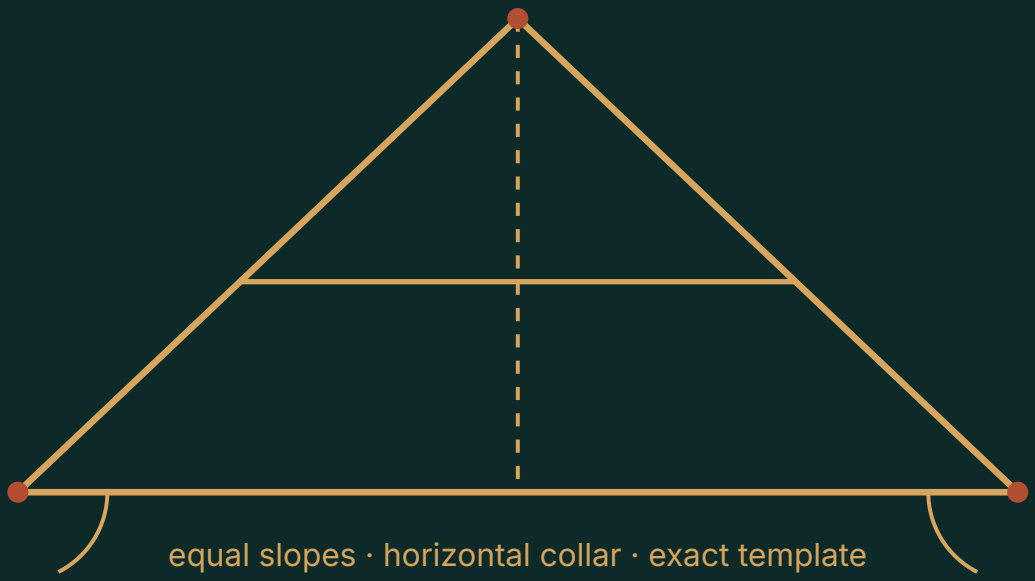
Lines, angles, triangles and constructions

A single straight line contains 180°, and a full turn contains 360°. From those two facts, and one statement about parallel lines, an entire system of shape reasoning follows. Constructions turn that reasoning into a repeatable physical procedure.



KEY CONCEPT · RELATIONSHIPS

RELATED · SPACE

RELATED · REPRESENTATION

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

A workshop needs a triangular roof truss with equal slopes and a horizontal collar beam. Which angles must the carpenter set, how can the shape be checked without measuring every angle, and how can a full-size template be reproduced exactly?

CHAPTER 7 · OPENER

Reasoning with straight lines

Geometry begins with a few reliable facts and becomes powerful when those facts are connected. In this chapter you will name angle relationships, explain triangle and polygon sums, and use equal-radius constructions to force a property rather than guess it.

ENGAGEMENT

Before cutting timber, a carpenter can test a design with a scale drawing. Which measurements are enough to guarantee the shape? Which checks protect against a drawing that only looks correct?

Inquiry questions

- **Factual:** What do angles on a straight line and around a point add to?
- **Conceptual:** Why must the two shorter sides of a triangle together exceed the longest side?
- **Debatable:** When is a construction reliable enough to build from?

Learning goals

- Use angle facts on lines, at points and with parallel lines.
- Reason with triangle and polygon angle sums.
- Apply the triangle inequality and regular-polygon facts.
- Construct perpendicular bisectors, angle bisectors and SSS triangles.

MYP lens

Key concept: Relationships

Related concepts: Space, Representation

Global context: Orientation in space and time

Criteria: A, B, C and D in separate tasks

ATL FOCUS. In order for a student to think creatively and critically, students must choose an appropriate geometric fact or construction, test a design and explain why the method is reliable.

Language bank: acute · right · obtuse · reflex · vertically opposite · corresponding · alternate · co-interior · isosceles · exterior angle · polygon · perpendicular bisector · angle bisector · construction

CHAPTER 7 · START HERE

Ready check and chapter map

Use the ready check to find the ideas you already control. Then follow the map from concrete angle facts to constructions and design decisions.

Ready check

Before starting, check that you can: name acute, right and obtuse angles; draw and label an angle ABC; use a ruler to draw a 6 cm segment; open compasses to a given radius. If any skill is uncertain, ask for a quick demonstration before beginning 7.1.

Chapter map

1. **7.1 Lines, angles and angle facts** — name the relationship first: straight line, full turn, vertically opposite or right angle.
2. **7.2 Parallel lines and triangles** — use corresponding, alternate and co-interior angles only when the lines are parallel.
3. **7.3 Polygons and the triangle inequality** — split polygons into triangles; test whether three lengths can close a triangle.
4. **7.4 Ruler-and-compass constructions** — use equal radii to force a property, keep arcs visible and check the defining relationship.

EXPLAIN HOW YOU KNOW

For every answer, be ready to give a method, a diagram or a reason. A correct result without a check is only half an explanation.

CHAPTER 7 · HOW TO USE THE PRACTICE

Six kinds of mathematical thinking

The practice is organised by concept, then by the kind of thinking it develops. The labels describe the task, not a separate mark scheme.

Understand

Identify, classify, state, complete and interpret definitions, notation and properties.

Fluency

Calculate, solve, complete diagrams and use procedures accurately.

Reasoning

Justify, compare methods, find errors and explain why a statement is true or false.

Investigation

Observe, organise evidence, make conjectures, test them and draw conclusions.

Application

Use mathematics in design, measurement and everyday contexts, then interpret the result.

Challenge

Combine ideas, consider alternatives and explain deeper relationships.

WORKING SPACE

Use the space provided for diagrams, equations and checks. Construction tasks ask you to draw in your notebook at the stated size; printed figures are models, not always full-size templates.

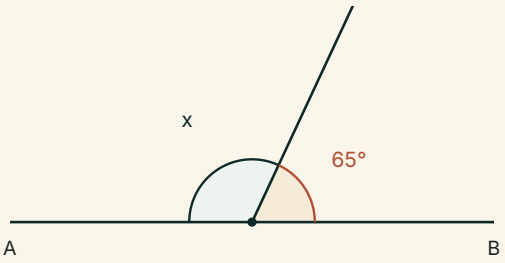
ATL EVIDENCE. As you work, record one place where you chose a method, one place where you tested a result, and one place where you explained why the method is reliable.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

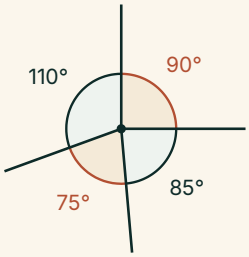
7.1 Lines, angles and angle facts

Name the relationship first: straight line, full turn, vertically opposite or right angle.

An angle measures a turn. A half-turn along a straight line is 180°, so adjacent angles on a straight line are supplementary. A full turn about a point is 360°. When two straight lines cross, vertically opposite angles are equal.



Straight line: the two angles complete 180°.



Full turn: the four angles complete 360°.

Key idea

Straight line: 180°. Full turn: 360°. Vertically opposite: equal. Right angle: 90°. Complementary angles sum to 90°; supplementary angles sum to 180°. The middle letter of angle ABC names the vertex B.

COMMON ERROR

Do not swap the totals: a straight line is 180°, a full turn is 360°. Do not treat vertically opposite angles as supplementary; they are equal. The adjacent pair is supplementary.

CHAPTER 7.1 · WORKED EXAMPLE 7.1.1

Missing angle on a straight line

PROBLEM

Two adjacent angles on a straight line are 47° and x. Find x.

VISUAL REPRESENTATION

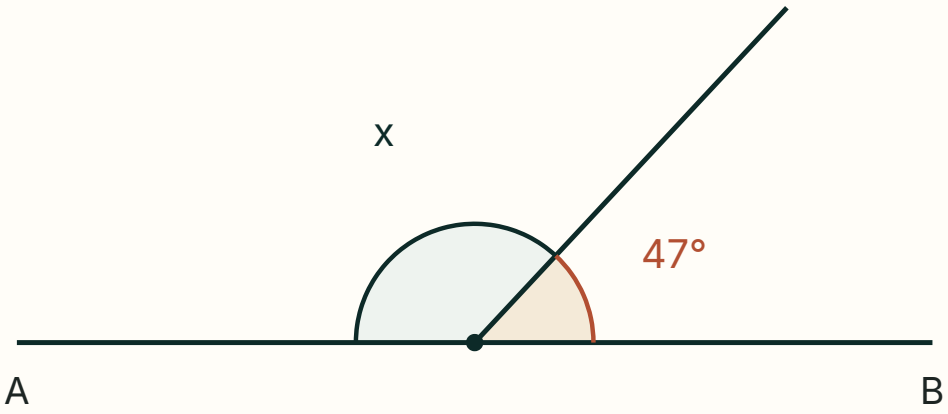


Figure 7.1 · The two angles share a vertex and lie on a straight line.

MATHEMATICAL REASONING

A straight line is a half-turn, so adjacent angles on it are supplementary: their sum is 180°.

STEP-BY-STEP SOLUTION

1. Identify the relationship: the two angles lie on one straight line.
2. Write the equation: 47° + x = 180°.
3. Subtract 47° from both sides: x = 180° − 47° = 133°.
4. Check: 47° + 133° = 180°.

Final answer. x = 133°.

Interpretation. The unknown angle is obtuse. The diagram and the equation agree: the two angles complete a half-turn.

CHAPTER 7.1 · WORKED EXAMPLE 7.1.2

Angles around a point

PROBLEM

Four angles meet at a point: 95°, 120°, 85° and x. Find x.

VISUAL REPRESENTATION

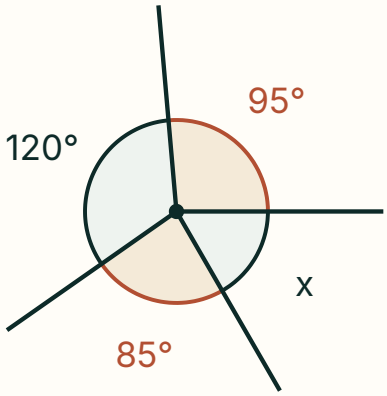


Figure 7.2 · Four sectors fill one full turn.

MATHEMATICAL REASONING

A full turn around a point is 360°. The four angles must fill the turn exactly.

STEP-BY-STEP SOLUTION

1. Add the known angles: 95° + 120° + 85° = 300°.
2. Subtract from the full turn: x = 360° − 300° = 60°.
3. Check: 95° + 120° + 85° + 60° = 360°.

Final answer. x = 60°.

Interpretation. The missing sector is acute. The visual model shows why the total must be 360°, not 180°.

CHAPTER 7.1 · WORKED EXAMPLE 7.1.3

Vertically opposite angles with algebra

PROBLEM

Two lines cross. Vertically opposite angles are labelled $3x + 10°$ and $x + 50°$. Find x and one adjacent angle.

VISUAL REPRESENTATION

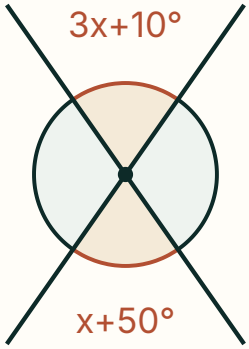


Figure 7.3 · Opposite angles are equal; adjacent angles sum to 180°.

MATHEMATICAL REASONING

Vertically opposite angles are equal. Adjacent angles on a straight line are supplementary.

STEP-BY-STEP SOLUTION

1. Set the opposite angles equal: $3x + 10 = x + 50$.
2. Collect terms: $2x = 40$, so $x = 20°$.
3. Each marked angle is $3(20) + 10 = 70°$.
4. An adjacent angle is $180° − 70° = 110°$. Check: $70° + 110° = 180°$.

Final answer. $x = 20°$; marked angles $70°$; adjacent angle $110°$.

Interpretation. The algebra confirms the visual symmetry: opposite sectors match, and neighbouring sectors complete a line.

CHAPTER 7.1 · PRACTICE

Angle language and turn facts

Practice 7.1 · Angle language and turn facts

7.1.01–7.1.03

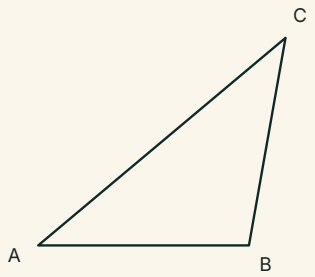
Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.1.01
UNDERSTAND

Classify each angle as acute, right, obtuse, straight or reflex: 35°, 90°, 125°, 180°, 210°. State the boundary values you used.

7.1.02
UNDERSTAND

In angle ABC, which point is the vertex? Which two rays form the angle? Draw a labelled sketch of any obtuse angle ABC.



7.1.03
UNDERSTAND

Complete the fact table and add one labelled sketch for each fact.

CONFIGURATION	FACT
Angles on a straight line	sum = ___°
Angles around a point	sum = ___°
Vertically opposite angles	they are ___

CHAPTER 7.1 · PRACTICE

Angle language and turn facts

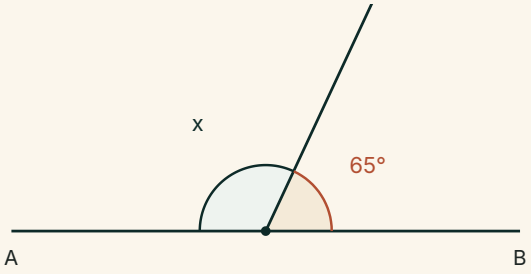
Practice 7.1 · Angle language and turn facts

7.1.04–7.1.05

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

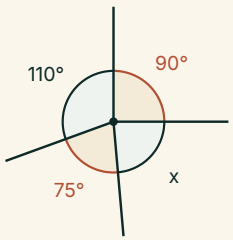
7.1.04
FLUENCY

Find x. State the angle fact used and check by addition.



7.1.05
FLUENCY

Three angles around a point are 90°, 110° and 75°. Find the fourth angle and check the full turn.



CHAPTER 7.1 · PRACTICE

Angle language and turn facts

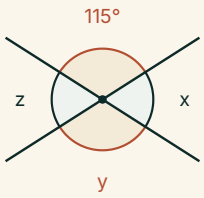
Practice 7.1 · Angle language and turn facts

7.1.06–7.1.07

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

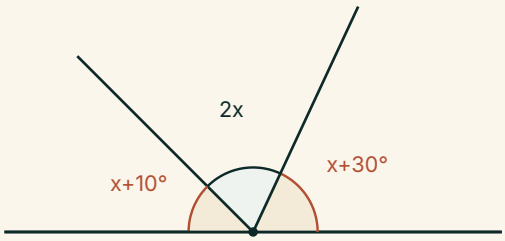
7.1.06
FLUENCY

Two lines cross. One angle is 115°. Find x, y and z, naming the relationship used for each.



7.1.07
FLUENCY

Three adjacent angles on a straight line are x + 30°, 2x and x + 10°. Find x and all three angles.



CHAPTER 7.1 · PRACTICE

Angle language and turn facts

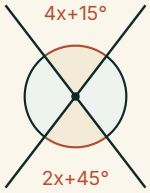
Practice 7.1 · Angle language and turn facts

7.1.08–7.1.10

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

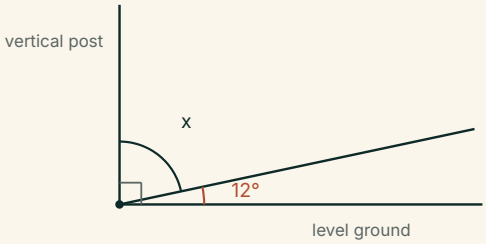
7.1.08 FLUENCY

The vertically opposite angles $4x + 15°$ and $2x + 45°$ are marked. Find x, the marked angles and one adjacent angle.



7.1.09 APPLICATION

A ramp makes $12°$ with level ground. A vertical post meets the ground at the same point, and the ramp lies between the ground and the post. Find the angle between the ramp and the post, and explain why the ground–post angle is $90°$.



7.1.10 APPLICATION

A turning sensor records three consecutive turns of $118°$, $97°$ and $85°$ before completing one full turn. Find the fourth turn. Would the answer change if the turns were recorded in a different order?

CHAPTER 7.1 · PRACTICE

Angle language and turn facts

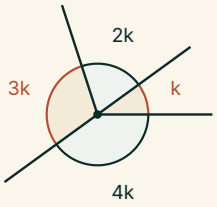
Practice 7.1 · Angle language and turn facts

7.1.11–7.1.13

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.1.11 APPLICATION

A logo divides a full turn into four angles in the ratio 1 : 2 : 3 : 4. Find the four angles. Is any pair complementary?



7.1.12 REASONING

A student writes two claims. Identify the incorrect claim, correct it, and draw a counterexample to the original claim.

CLAIM	STATEMENT
1	Adjacent angles on a straight line sum to 180°.
2	Angles around a point sum to 180°.

7.1.13 REASONING

True or false: if two adjacent angles on a straight line are equal, each must be 90°. Justify with an equation and a sketch.

CHAPTER 7.1 · PRACTICE

Angle language and turn facts

Practice 7.1 · Angle language and turn facts

7.1.14–7.1.16

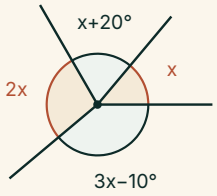
Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.1.14 REASONING

Complete the missing steps: for angles 2x, x + 40° and 80° around a point, __ + __ + __ = 360°; so __ x = __; x = __. Check the three angles.

7.1.15 CHALLENGE

Four angles around a point are x, x + 20°, 2x and 3x − 10°. Find x and the largest angle. Explain why the largest angle is not reflex.



7.1.16 CHALLENGE

Draw two different pairs of crossing lines with one angle 70°. Label all four angles in each diagram. Compare your diagrams with a partner: what stays the same and what changes?

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.2 Parallel lines and transversals

A transversal crosses two or more lines. When the crossed lines are parallel, the angles fall into equal or supplementary families.

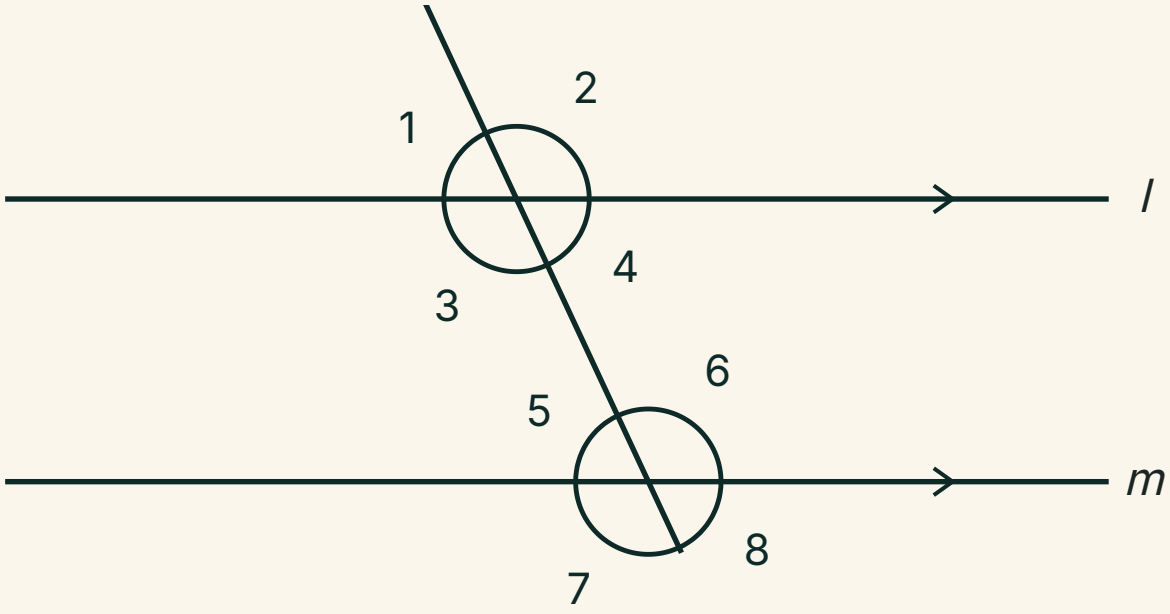


Figure 7.4 · With l parallel to m: corresponding and alternate angles are equal; co-interior angles are supplementary.

Key idea

Corresponding angles sit in the same position at each intersection. Alternate angles lie on opposite sides of the transversal between the parallel lines. Co-interior angles lie on the same side between the parallel lines.

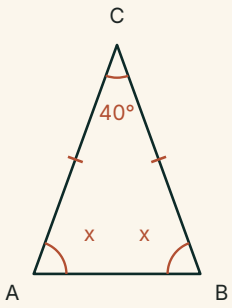
WATCH FOR THIS MISCONCEPTION

Do not assume lines are parallel or angles equal unless marked or stated. Conversely, equal alternate angles or supplementary co-interior angles can show that two lines are parallel.

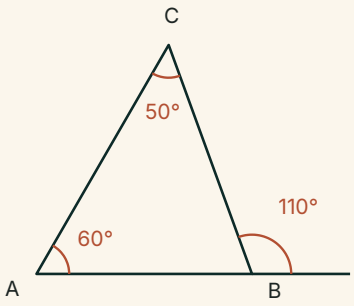
CHAPTER 7 · MATHEMATICAL DEVELOPMENT

Triangle and exterior-angle facts

Every triangle has angle sum 180°. An exterior angle equals the sum of the two opposite interior angles.



Equal sides give equal base angles.



Exterior angle = sum of opposite interior angles.

Key idea

Triangles may be classified by sides (equilateral, isosceles, scalene) and by angles (acute, right, obtuse). In an isosceles triangle the angles opposite the equal sides are equal; conversely, equal angles face equal sides. In a right triangle the two acute angles are complementary.

COMMON ERROR

The exterior angle is equal to the sum of the two opposite interior angles. It is not added to them to make 180°. The adjacent interior angle is the one supplementary to the exterior angle.

CHAPTER 7.2 · WORKED EXAMPLE 7.2.1

Parallel-line angle chase

PROBLEM

Lines l and m are parallel. The marked angle is 68°. Find a, b and c.

VISUAL REPRESENTATION

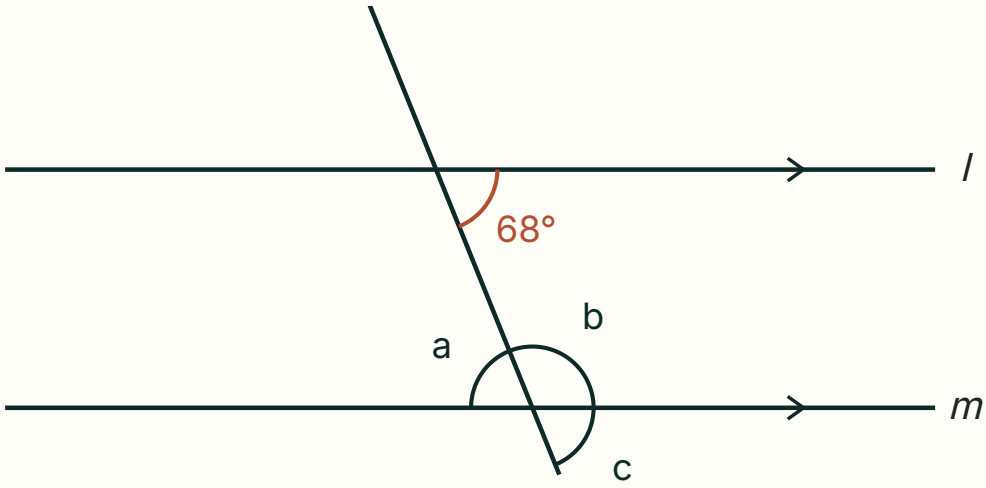


Figure 7.5 · Parallel marks justify the angle relationships.

MATHEMATICAL REASONING

With parallel lines, alternate angles are equal, co-interior angles are supplementary, and corresponding angles are equal.

STEP-BY-STEP SOLUTION

1. a is alternate to 68°, so a = 68°.
2. b is co-interior with 68°, so b = 180° − 68° = 112°.
3. c corresponds to the 68° angle (both are in the lower-right position), so c = 68°.
4. Check: a + b = 68° + 112° = 180°.

Final answer. a = 68°, b = 112°, c = 68°.

Interpretation. The transversal creates two angle families: 68° and 112°. Naming the relationship prevents guessing from appearance.

CHAPTER 7.2 · WORKED EXAMPLE 7.2.2

Exterior angle of a triangle

PROBLEM

An exterior angle of a triangle is 115°. One opposite interior angle is 48°. Find the other opposite interior angle and the adjacent interior angle.

VISUAL REPRESENTATION

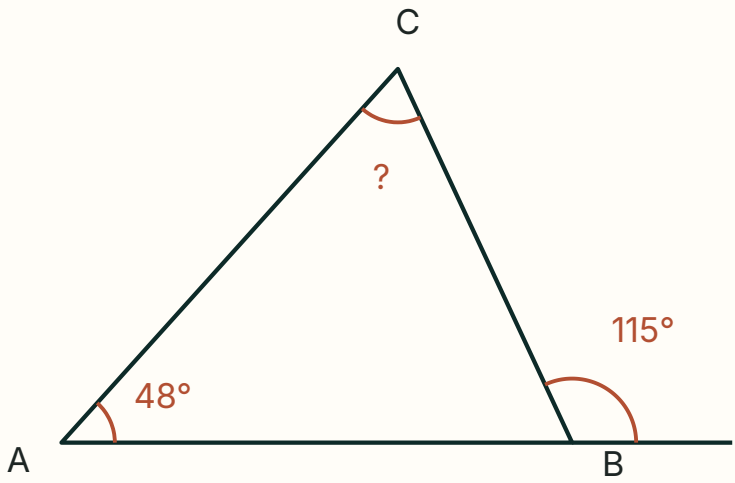


Figure 7.6 · The exterior angle is outside the triangle at the extended base.

MATHEMATICAL REASONING

An exterior angle equals the sum of the two opposite interior angles. It is supplementary to the interior angle at the same vertex.

STEP-BY-STEP SOLUTION

1. Other opposite interior angle = 115° − 48° = 67°.
2. Adjacent interior angle = 180° − 115° = 65°.
3. Check the triangle sum: 48° + 67° + 65° = 180°.

Final answer. Other opposite angle 67°; adjacent interior angle 65°.

Interpretation. The exterior angle is larger than either opposite interior angle. The diagram shows which angles belong to the exterior-angle relationship.

CHAPTER 7.2 · WORKED EXAMPLE 7.2.3

Isosceles base angles

PROBLEM

An isosceles triangle has apex angle 40°. Find each base angle.

VISUAL REPRESENTATION

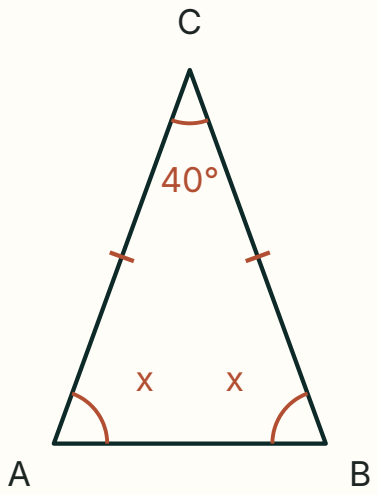


Figure 7.7 · Equal sides force equal base angles.

MATHEMATICAL REASONING

Equal sides have equal opposite angles. The triangle angle sum is 180°.

STEP-BY-STEP SOLUTION

1. The tick marks show two equal sides, so the base angles are equal. Call each x.
2. Write $2x + 40° = 180°$.
3. Then $2x = 140°$, so $x = 70°$.
4. Check: $70° + 70° + 40° = 180°$.

Final answer. Each base angle is 70°.

Interpretation. The equal-side marks are the evidence for equal base angles; the calculation then uses the angle sum.

CHAPTER 7.2 · WORKED EXAMPLE 7.2.4

Why every triangle has angle sum 180°

PROBLEM

Use a line through the apex parallel to the base to explain why the angles of a triangle sum to 180°.

VISUAL REPRESENTATION

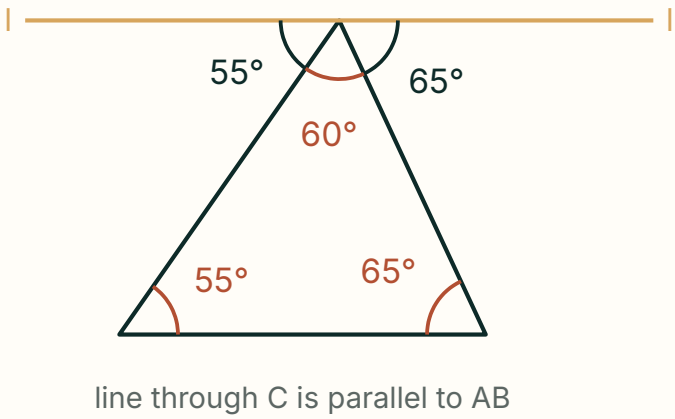


Figure 7.8 · The parallel line transfers the base angles to the apex.

MATHEMATICAL REASONING

Alternate angles between parallel lines are equal. Angles on a straight line sum to 180°.

STEP-BY-STEP SOLUTION

1. Draw a line through the apex parallel to the base.
2. The left base angle equals the left angle at the apex (alternate angles). The right base angle equals the right angle at the apex.
3. The three angles on the straight line through the apex are therefore the three triangle angles.
4. They sum to 180°, so the triangle angles sum to 180°.

Final answer. A + B + C = 180° for every triangle.

Interpretation. This is a visual argument, not a measurement of one triangle. It explains why the rule is general.

CHAPTER 7.2 · PRACTICE

Parallel lines and transversals

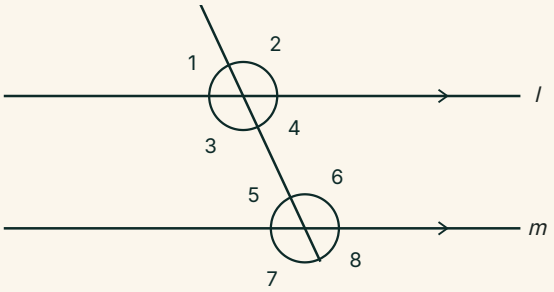
Practice 7.2 · Parallel lines and transversals

7.2.01–7.2.02

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

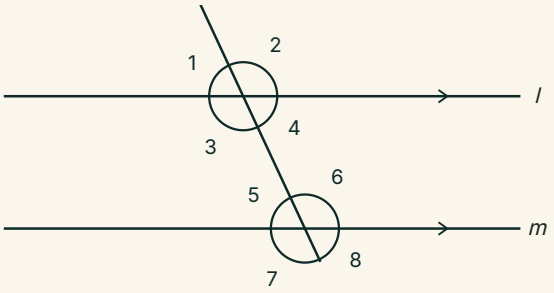
7.2.01
UNDERSTAND

Use the numbered diagram. Name one pair of corresponding angles, one pair of alternate angles and one pair of co-interior angles. State which pairs are equal and which are supplementary.



7.2.02
UNDERSTAND

True or false: any two angles between parallel lines are equal. Use the diagram to justify your answer.



CHAPTER 7.2 · PRACTICE

Parallel lines and transversals

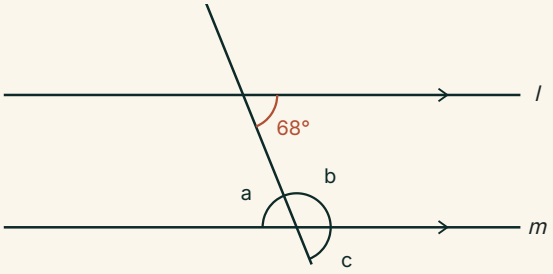
Practice 7.2 · Parallel lines and transversals

7.2.03–7.2.05

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

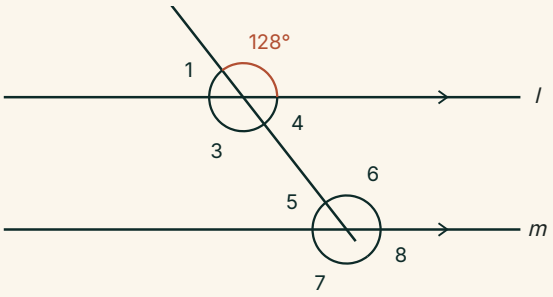
7.2.03 FLUENCY

Find a, b and c. Give a reason for each value.



7.2.04 FLUENCY

In the diagram, angle 2 = 128°. Find angle 5, angle 6 and angle 7, naming each relationship.



7.2.05 REASONING

Two lines are cut by a transversal. A pair of co-interior angles measures 112° and 68°. Are the lines parallel? Justify. Then draw a non-parallel pair and explain why the same conclusion cannot be drawn from appearance alone.

CHAPTER 7.2 · PRACTICE

Parallel lines and transversals

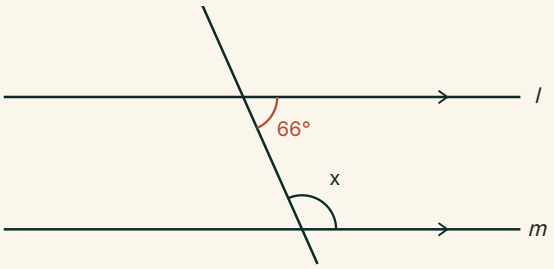
Practice 7.2 · Parallel lines and transversals

7.2.11–7.2.19

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

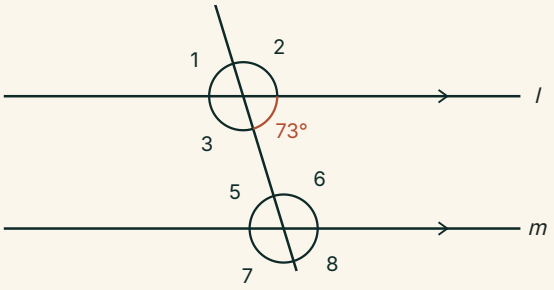
7.2.11 APPLICATION

A collar is parallel to the base. At a rafter joint the co-interior angle inside the frame is 66°. Find the angle x on the same side of the transversal at the other joint.



7.2.19 APPLICATION

Two parallel shelves are crossed by a support. One acute angle is 73°. Complete a table showing all eight angles, then state how many are acute and how many are obtuse.



CHAPTER 7.2 · PRACTICE

Parallel lines and transversals

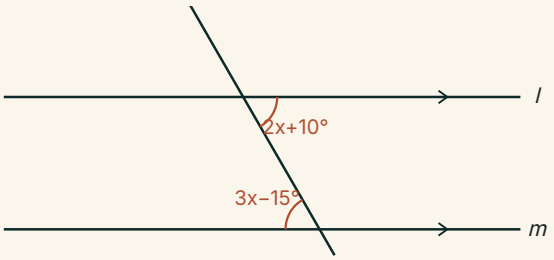
Practice 7.2 · Parallel lines and transversals

7.2.20–7.2.20

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.2.20
REASONING

The alternate angles marked $2x + 10°$ and $3x - 15°$ are formed by parallel lines. Find x, the marked angles, and the co-interior partner of one marked angle.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

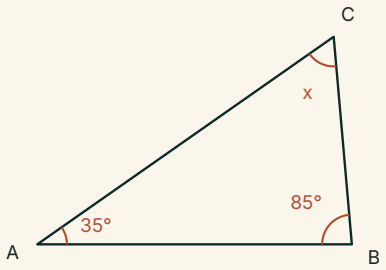
Practice 7.2 · Triangle and exterior-angle facts

7.2.06–7.2.07

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

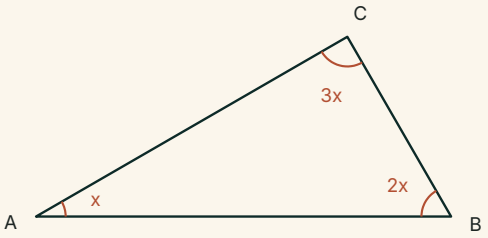
7.2.06
UNDERSTAND

A triangle has angles 35° and 85°. Find the third angle and classify the triangle by angles and by sides.



7.2.07
FLUENCY

The angles of a triangle are x, 2x and 3x. Find x and classify the triangle.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

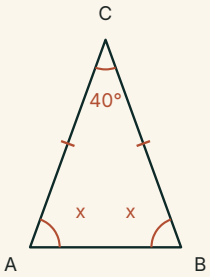
Practice 7.2 · Triangle and exterior-angle facts

7.2.08–7.2.09

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

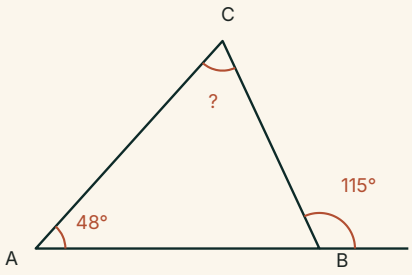
7.2.08
FLUENCY

Find each base angle x. Which marks tell you the base angles are equal?



7.2.09
FLUENCY

An exterior angle is 115° and one opposite interior angle is 48°. Find the other opposite interior angle and the interior angle adjacent to the exterior angle.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

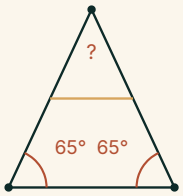
Practice 7.2 · Triangle and exterior-angle facts

7.2.10–7.2.12

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.2.10 APPLICATION

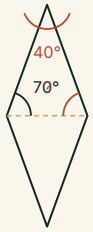
The truss has equal base angles of 65°. Find the apex angle and the exterior angle at a base. Classify the triangle and state what this implies about the rafters.



8 m base

7.2.12 APPLICATION

The kite is made from two identical isosceles triangles joined along their bases. The upper triangle has apex 40°. Find the marked base angle, then find the whole angle at the left tip of the kite.



upper and lower triangles are identical

CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

Practice 7.2 · Triangle and exterior-angle facts

7.2.13–7.2.15

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.2.13 APPLICATION

A right-triangular ramp frame has a 14° angle with the ground. A handrail continues the sloping side beyond the top. Find the top interior angle and the exterior angle between the handrail extension and the vertical post.

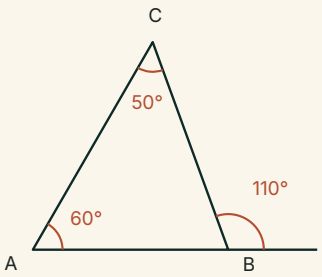


7.2.14 REASONING

Can a triangle have two obtuse angles? Explain using the angle sum, and sketch what goes wrong if you try.

7.2.15 REASONING

A student says the exterior angle 110° should be added to the opposite interior angles 60° and 50° to make 180°. Identify the error, correct it, and find the adjacent interior angle.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

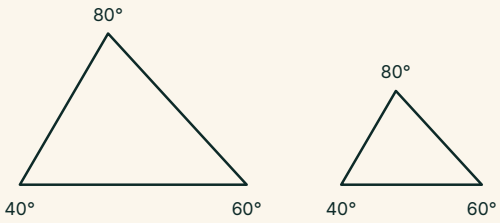
Practice 7.2 · Triangle and exterior-angle facts

7.2.16–7.2.17

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

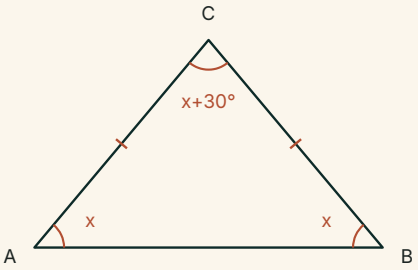
7.2.16
REASONING

Compare the two triangles. What is the same and what is different? Does having the same three angles force the side lengths to be the same? Justify.



7.2.17
CHALLENGE

An isosceles triangle has base angles x and apex x + 30°. Find all three angles. Show two different checks.



CHAPTER 7.2 · PRACTICE

Triangle and exterior-angle facts

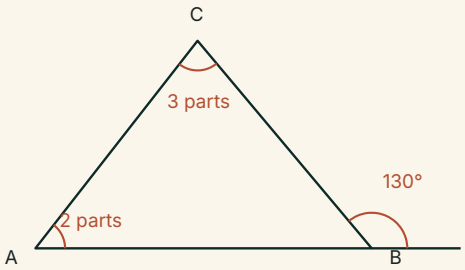
Practice 7.2 · Triangle and exterior-angle facts

7.2.18–7.2.18

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.2.18
CHALLENGE

An exterior angle of a triangle is 130°. The two opposite interior angles are in the ratio 2 : 3. Find all three interior angles and classify the triangle.



CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.3 Polygons and their angles

A polygon is a closed plane shape made from straight sides. Its interior angle sum depends on the number of sides.

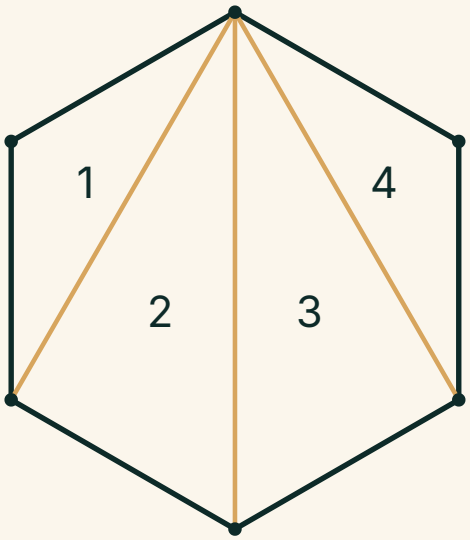


Figure 7.9 · A hexagon splits into four triangles from one vertex.

Choose one vertex and draw diagonals to every non-adjacent vertex. An n-sided polygon divides into n−2 triangles, so its interior angle sum is (n−2)×180°. The exterior angles of any convex polygon sum to 360°.

Key idea

Interior sum: (n−2)×180°. Exterior sum: 360°. Regular polygon: each exterior = 360°÷n; each interior = 180°−360°÷n.

CHAPTER 7.3 · WORKED EXAMPLE 7.3.1

Missing angle in a quadrilateral

PROBLEM

A quadrilateral has angles 85°, 95°, 110° and x. Find x.

VISUAL REPRESENTATION

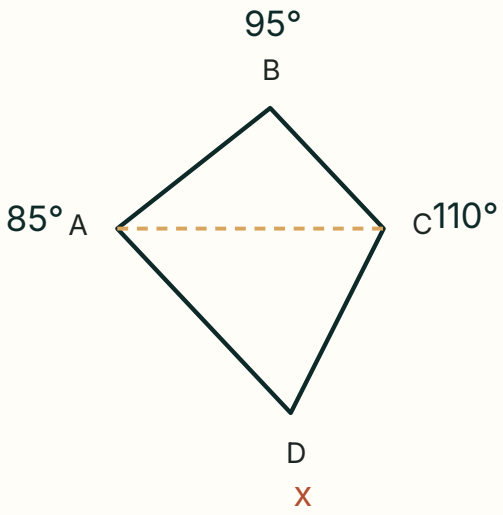


Figure 7.10 · One diagonal divides the quadrilateral into two triangles.

MATHEMATICAL REASONING

A quadrilateral can be split into two triangles, so its interior angle sum is $2 \times 180° = 360°$.

STEP-BY-STEP SOLUTION

1. Add the known angles: $85° + 95° + 110° = 290°$.
2. Subtract from 360°: $x = 360° - 290° = 70°$.
3. Check: $85° + 95° + 110° + 70° = 360°$.

Final answer. x = 70°.

Interpretation. The diagonal in the diagram shows where the 360° comes from: two triangle sums.

CHAPTER 7.3 · WORKED EXAMPLE 7.3.2

Regular polygon from an exterior angle

PROBLEM

A regular polygon has exterior angle 40°. Find the number of sides and each interior angle.

VISUAL REPRESENTATION

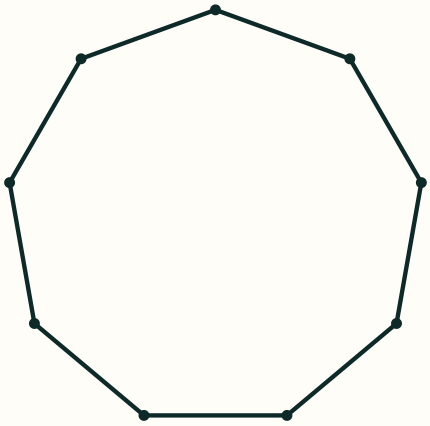


Figure 7.11 · A regular nonagon has nine equal exterior turns.

MATHEMATICAL REASONING

The exterior angles of any convex polygon sum to 360°. In a regular polygon they are equal. Interior and exterior angles at a vertex are supplementary.

STEP-BY-STEP SOLUTION

1. Number of sides $n = 360° \div 40° = 9$.
2. Each interior angle $= 180° - 40° = 140°$.
3. Check with the interior-sum formula: $(9 - 2) \times 180° = 1260°$, and $1260° \div 9 = 140°$.

Final answer. 9 sides; each interior angle 140°.

Interpretation. A small exterior angle means many sides. The polygon is a regular nonagon.

CHAPTER 7.3 · PRACTICE

Polygon angle sums

Practice 7.3 · Polygons

7.3.01–7.3.02

7.3.01 UNDERSTAND

Name the polygons and complete the number of sides. Then state what makes a polygon regular.

NAME	SIDES
Pentagon	___
Hexagon	___
Octagon	___

7.3.02 UNDERSTAND

Compare a regular hexagon with an irregular hexagon. Sketch each and label one property that is the same and one that may differ.

CHAPTER 7.3 · PRACTICE

Polygon angle sums

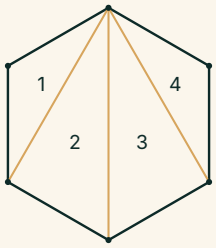
Practice 7.3 · Polygons

7.3.03–7.3.04

7.3.03
FLUENCY

Complete the triangulation table. Use the hexagon diagram to check the last row.

POLYGON	TRIANGLES	INTERIOR SUM
Quadrilateral	___	___°
Pentagon	___	___°
Hexagon	___	___°



7.3.04
FLUENCY

Find the interior angle sum of a heptagon, a nonagon and a decagon. Show the number of triangles first.

POLYGON	SIDES	TRIANGLES	SUM
Heptagon	7	___	___°
Nonagon	9	___	___°
Decagon	10	___	___°

CHAPTER 7.3 · PRACTICE

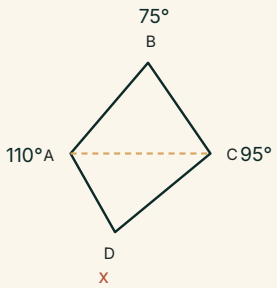
Polygon angle sums

Practice 7.3 · Polygons

7.3.05–7.3.06

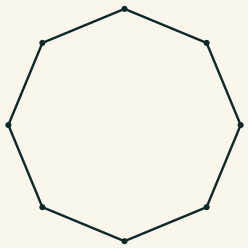
7.3.05
FLUENCY

Find the missing angle x in the quadrilateral. Check by adding all four angles.



7.3.06
FLUENCY

A regular polygon has exterior angle 45°. Find the number of sides and each interior angle. Sketch the polygon and mark one interior–exterior pair.



CHAPTER 7.3 · PRACTICE

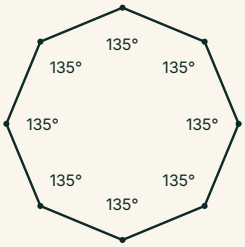
Polygon angle sums

Practice 7.3 · Polygons

7.3.07–7.3.08

7.3.07
APPLICATION

A floor-tile design uses a regular polygon whose interior angle is 135°. Identify the polygon. Can copies of this tile alone meet around a point with no gap? Justify.



7.3.08
REASONING

A student walks once around a convex polygon, turning through the exterior angle at each vertex. Complete the table and explain why the total turn is 360° for every convex polygon.

POLYGON	NUMBER OF EXTERIOR TURNS	TOTAL TURN
Triangle	3	___°
Quadrilateral	4	___°
Decagon	10	___°

CHAPTER 7 · INQUIRY LAUNCH

Which lengths close a triangle?

Three strips joined end to end may or may not close a triangle. Predict first, then test. Do not state a rule until the evidence forces it.

SIDE LENGTHS (CM)	PREDICTION	ARCS MEET ABOVE THE BASE?	TRIANGLE?
3, 4, 5			
3, 4, 8			
4, 5, 9			
6, 7, 10			
2, 2, 3			

OBSERVE

When the arcs meet on the base, the “triangle” has zero area. When they do not meet, there is a gap. What must be true about the two shorter lengths for a real apex to exist?

Keep this prediction. You will return to it after testing more cases.

CHAPTER 7 · INVESTIGATION · CRITERION B PRACTICE

Test lengths with compass arcs

Choose a longest side c and two shorter sides a and b. Set the base to c, then draw an arc of radius a from one endpoint and an arc of radius b from the other. Record what happens.

A, B, C (CM)	A+B	COMPARE A+B WITH C	ARCS MEET ABOVE BASE?	TRIANGLE?
4, 5, 6				
2, 3, 6				
5, 8, 13				
6, 8, 9				
your own case				

1. Complete the table. Include at least one borderline case of your own.
2. Compare a+b with c for each row. Write a conjecture that predicts whether a triangle can be made.
3. Test your conjecture on 5, 8, 12 and 7, 7, 14. Does it still work?
4. Explain why the arcs behave as they do. Use a labelled sketch.

ATL EVIDENCE. Choose your own extra case, organise the evidence, test the conjecture and explain the result.

CHAPTER 7 · INVESTIGATION · DISCOVERY

From evidence to a rule

Write your conjecture before comparing it with the formal statement below.

DISCOVERY

The two shorter sides must together be **strictly greater** than the longest side. If a+b=c, the arcs meet on the base and the triangle has zero area. If a+b<c, the arcs do not meet.

Conclusion

For sides a, b, c with c longest, a triangle exists if and only if a+b>c. The other two inequalities then hold automatically. This is the triangle inequality.

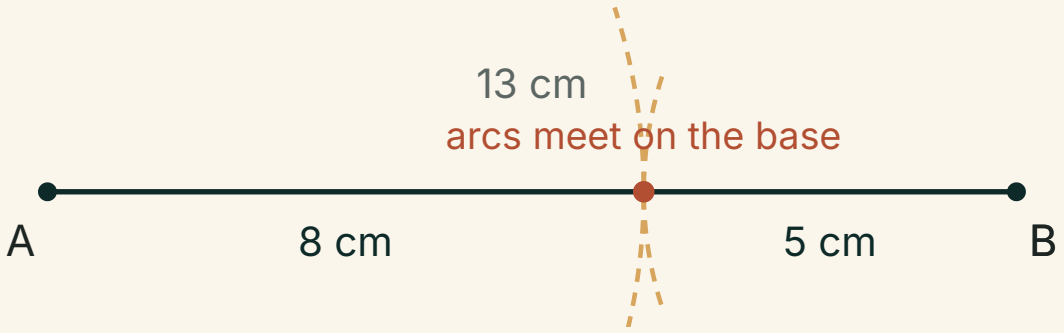


Figure 7.12 · Equality gives a degenerate straight line.

CHAPTER 7.3 · WORKED EXAMPLE 7.3.3

Possible third sides

PROBLEM

Two sides of a triangle are 7 cm and 11 cm. The third side is a whole number of centimetres. Find all possible values.

VISUAL REPRESENTATION

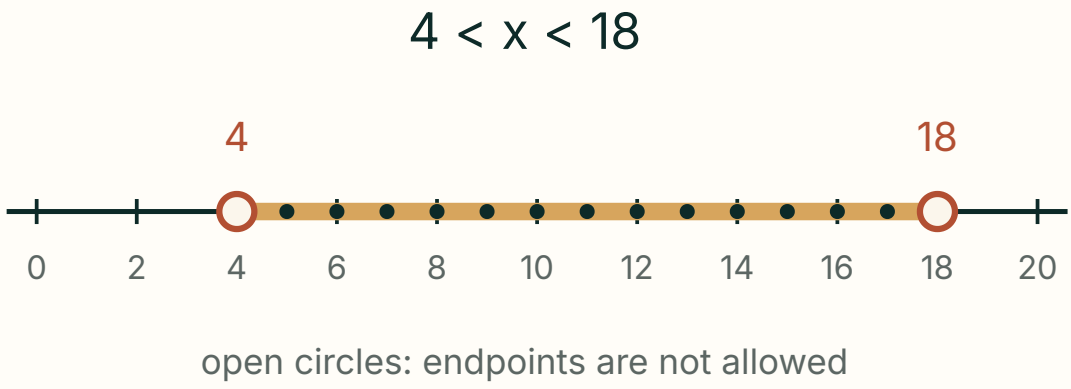


Figure 7.13 · The number line represents 4 < x < 18.

MATHEMATICAL REASONING

The third side must be greater than the difference of the other two sides and less than their sum. Endpoints give degenerate cases.

STEP-BY-STEP SOLUTION

1. Lower boundary: 11 − 7 = 4, so x > 4.
2. Upper boundary: 11 + 7 = 18, so x < 18.
3. Whole-number values are 5, 6, 7, ..., 17.
4. Check endpoints: x = 4 gives 4 + 7 = 11; x = 18 gives 7 + 11 = 18. Both are straight-line cases, not triangles.

Final answer. x = 5, 6, 7, ..., 17 cm.

Interpretation. The open circles on the number line show that 4 and 18 are excluded. The integer dots are the possible whole-centimetre lengths.

CHAPTER 7.3 · PRACTICE

Triangle inequality

Practice 7.3 · Possible triangles

7.3.09–7.3.11

7.3.09 INVESTIGATION

Test whether each triple can form a triangle. Record the sum of the two shorter sides and compare it with the longest side.

SIDES (CM)	SHORTER SUM	LONGEST	TRIANGLE?
3, 4, 8	___	___	___
5, 5, 9	___	___	___
6, 7, 10	___	___	___

7.3.10 INVESTIGATION

Complete the table. Include one row of your own that is a borderline case. What pattern separates possible triangles from impossible ones?

A, B, C (CM)	A+B	COMPARE WITH C	TRIANGLE?
4, 5, 6	___	___	___
2, 3, 6	___	___	___
5, 8, 13	___	___	___
6, 8, 9	___	___	___
your row	___	___	___

7.3.11 FLUENCY

A frame has sides 7 cm, 9 cm and 15 cm. Can it be a triangle? If not, change only the longest side to the smallest whole-number length that makes a triangle.

CHAPTER 7.3 · PRACTICE

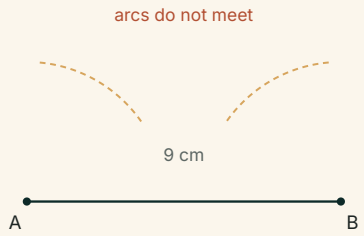
Triangle inequality

Practice 7.3 · Possible triangles

7.3.12–7.3.14

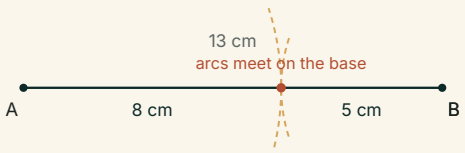
7.3.12
APPLICATION

A student says a triangular brace with sides 4 cm, 4 cm and 9 cm is possible because 4+9>4. Correct the reasoning and sketch what happens when the two 4 cm sides are joined to the 9 cm base.



7.3.13
REASONING

Use the arcs to explain why sides 5 cm, 8 cm and 13 cm do not form a triangle. What changes if the 13 cm side becomes 12 cm?



7.3.14
CHALLENGE

Two sides of a triangle are 7 cm and 11 cm. The third side is a whole number of centimetres. Find all possible values and represent them on a number line.

CHAPTER 7.3 · PRACTICE

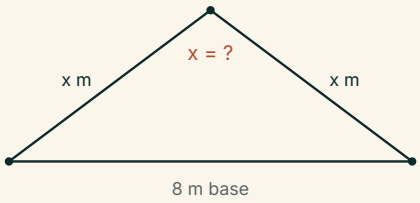
Triangle inequality

Practice 7.3 · Possible triangles

7.3.15–7.3.17

7.3.15 CHALLENGE

A truss model has an 8 m base and equal rafters of length x m. Find the smallest whole-number value of x that makes a triangle. Explain why x = 4 is not enough.

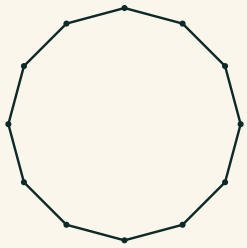


7.3.16 CHALLENGE

Use compass arcs to explain why the two shorter sides of a triangle must together exceed the longest side. Include the cases a+b<c, a+b=c and a+b>c in your explanation.

7.3.17 CHALLENGE

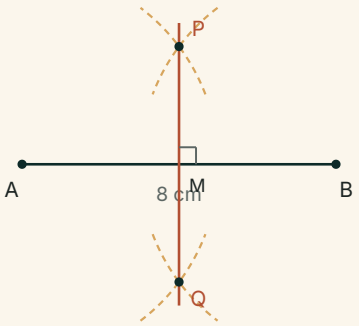
A regular polygon has each interior angle five times its exterior angle. Find the number of sides. Show the relationship between one interior and exterior pair.



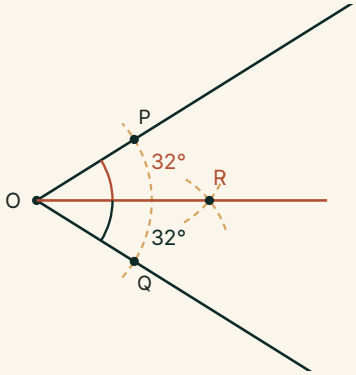
CHAPTER 7 · MATHEMATICAL DEVELOPMENT

7.4 Constructions with ruler and compass

A construction is an argument made with instruments. Each arc records a fixed distance from a named point.



Perpendicular bisector: equidistant from two endpoints.



Angle bisector: equidistant from the two arms.

CONSTRUCTION	COMPASS SETTING	HOW TO TEST
Perpendicular bisector of AB	One radius greater than half AB, unchanged from A and B	Crosses AB at its midpoint and makes 90°
Angle bisector	Arc from vertex cuts both arms; same radius from those points	Two half-angles are equal
Triangle from three sides	Base equal to one side; radii equal to the other two sides	Arcs meet; two shorter sides exceed the longest

COMMON ERROR

Never change the compass between a pair of arcs. Check the defining property, not the appearance of the drawing.

CHAPTER 7 · MATHEMATICAL DEVELOPMENT

Constructing triangles from sides

To construct a triangle from three sides, fix the base and locate the apex with two arcs.

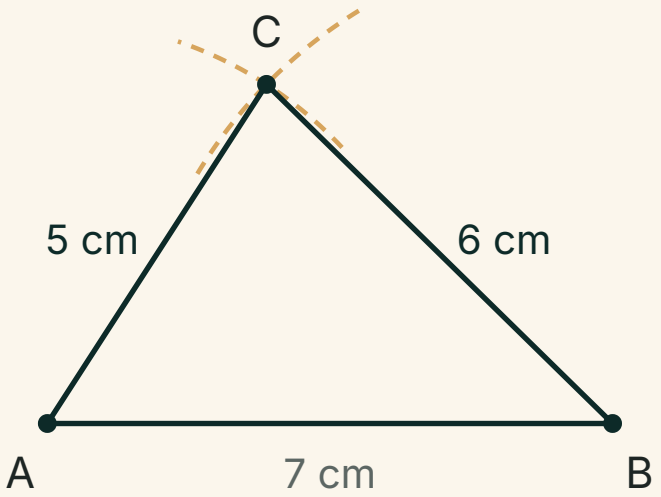


Figure 7.14 · The arc intersection fixes the third vertex.

The triangle inequality predicts whether the arcs meet above the base. If the two shorter sides together equal the longest side, the apex lies on the base; if they are shorter, the arcs do not meet.

Key idea

SSS construction: base AB, arc from A with radius AC, arc from B with radius BC. The intersection is C. Keep the arcs visible as evidence.

CHAPTER 7.4 · WORKED EXAMPLE 7.4.1

Perpendicular bisector

PROBLEM

Construct the perpendicular bisector of AB = 8 cm. State the midpoint length and the check.

VISUAL REPRESENTATION

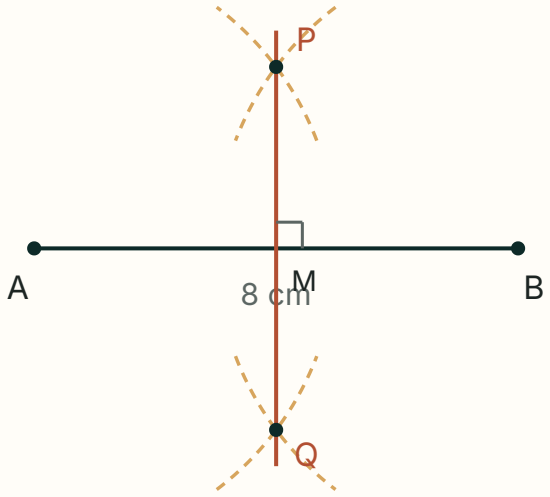


Figure 7.15 · The same compass radius is used from both endpoints.

MATHEMATICAL REASONING

Points on the perpendicular bisector are equidistant from A and B. Equal-radius arcs locate two such points.

STEP-BY-STEP SOLUTION

1. Draw AB = 8 cm.
2. Choose one radius greater than half of AB, for example 5 cm. Keep it fixed.
3. Draw arcs from A and from B above and below AB. Label the intersections P and Q.
4. Draw PQ. It crosses AB at its midpoint M, so AM = MB = 4 cm, and PQ is perpendicular to AB.

Final answer. AM = MB = 4 cm; PQ is perpendicular to AB.

Interpretation. The arcs are evidence: P and Q are each the same distance from A and B, so the line through them is the perpendicular bisector.

CHAPTER 7.4 · WORKED EXAMPLE 7.4.2

Angle bisector

PROBLEM

Bisect a 64° angle with ruler and compasses. Find each half.

VISUAL REPRESENTATION

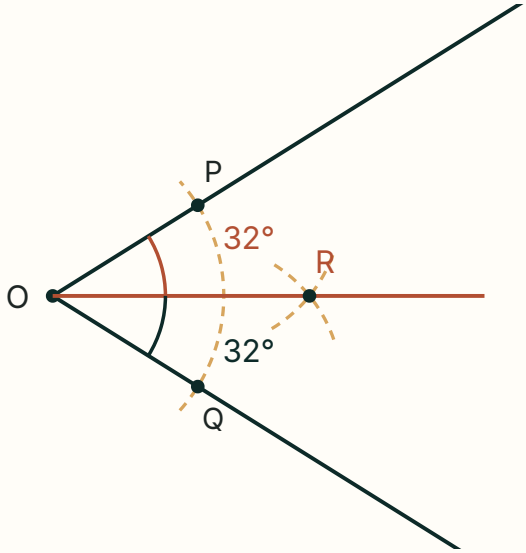


Figure 7.16 · The bisector is the ray OR.

MATHEMATICAL REASONING

An angle bisector divides an angle into two equal angles. Equal-radius arcs from the arms locate a point on the bisector.

STEP-BY-STEP SOLUTION

1. Draw an arc from the vertex O, cutting the arms at P and Q.
2. With the same radius from P and Q, draw arcs meeting at R.
3. Draw OR. This is the angle bisector.
4. Each half is $64° \div 2 = 32°$. Check: $32° + 32° = 64°$.

Final answer. Each half is 32°.

Interpretation. The construction forces equality; measuring is only a check. Do not confuse this with the perpendicular bisector of a segment.

CHAPTER 7.4 · WORKED EXAMPLE 7.4.3

Triangle from three sides (SSS)

PROBLEM

Construct a triangle with sides 5 cm, 6 cm and 7 cm.

VISUAL REPRESENTATION

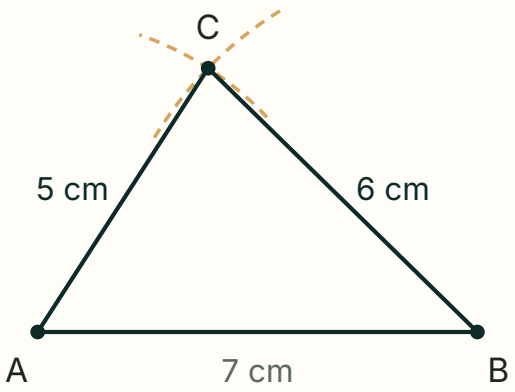


Figure 7.17 · The arc intersection fixes the third vertex.

MATHEMATICAL REASONING

If two sides and the base are fixed, the third vertex is the intersection of two circles/arcs. The triangle inequality predicts whether the arcs meet.

STEP-BY-STEP SOLUTION

1. Check: 5 + 6 > 7, so a triangle is possible.
2. Draw base AB = 7 cm.
3. From A draw an arc of radius 5 cm; from B draw an arc of radius 6 cm.
4. Label the intersection C and join AC and BC. Keep the arcs visible.

Final answer. Triangle ABC has AB = 7 cm, AC = 5 cm and BC = 6 cm.

Interpretation. The arcs locate the only possible apex on each side of the base. The two choices are mirror images.

CHAPTER 7.4 · WORKED EXAMPLE 7.4.4

Isosceles triangle and a locus

PROBLEM

Construct an isosceles triangle with equal sides 6 cm and base 8 cm. Explain why the perpendicular bisector of the base passes through the apex.

VISUAL REPRESENTATION

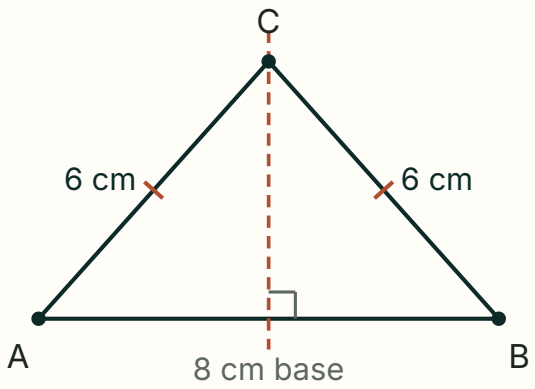


Figure 7.18 · Equal sides place the apex on the perpendicular bisector.

MATHEMATICAL REASONING

The apex is equidistant from the base endpoints. The perpendicular bisector is the locus of all points equidistant from the endpoints.

STEP-BY-STEP SOLUTION

1. Draw base AB = 8 cm.
2. With radius 6 cm from A and from B, draw arcs meeting at C.
3. Join AC and BC. Then AC = BC = 6 cm.
4. Because C is equidistant from A and B, it lies on the perpendicular bisector of AB.

Final answer. The perpendicular bisector of AB passes through C.

Interpretation. The equal sides are not just a picture feature; they place the apex on a precise locus. This connects construction with reasoning.

CHAPTER 7.4 · PRACTICE

Bisectors and loci

Practice 7.4 · Bisectors and loci

7.4.01–7.4.03

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

7.4.01
UNDERSTAND

Select the construction that locates all points equally distant from two points A and B: angle bisector, perpendicular bisector, parallel line, or circle centre A. Contrast this with the locus of points inside the angle between two intersecting roads that are equally distant from the roads.

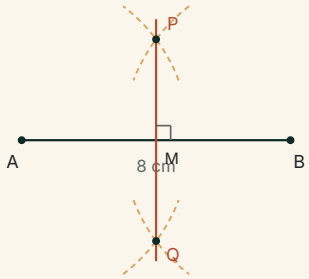
GIVEN	LOCUS NEEDED
Two points A and B	___
Inside the angle between two roads	___

7.4.02
UNDERSTAND

In a ruler-and-compass construction, what do equal-radius arcs record? Why must the compass setting stay fixed between a pair of arcs?

7.4.03
FLUENCY

Order the steps for a perpendicular bisector of AB = 8 cm: (i) draw arcs from A and B with the same radius; (ii) draw AB; (iii) join the arc intersections; (iv) open the compass to more than half AB. State the radius condition.



CHAPTER 7.4 · PRACTICE

Bisectors and loci

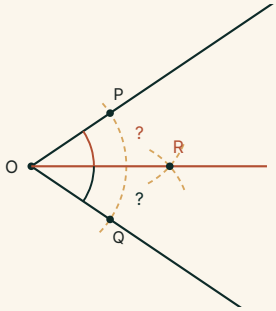
Practice 7.4 · Bisectors and loci

7.4.04–7.4.05

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

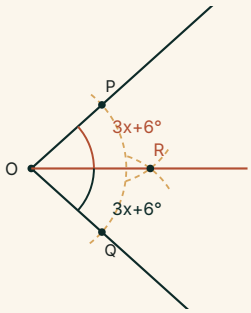
7.4.04
FLUENCY

An angle of 68° is bisected. Find each half and describe how the arcs show the two halves are equal.



7.4.05
FLUENCY

An 84° angle is bisected. Each half is labelled 3x + 6°. Find x and check.



CHAPTER 7.4 · PRACTICE

Bisectors and loci

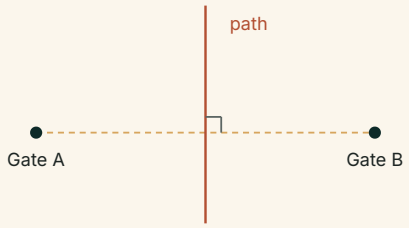
Practice 7.4 · Bisectors and loci

7.4.06–7.4.08

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

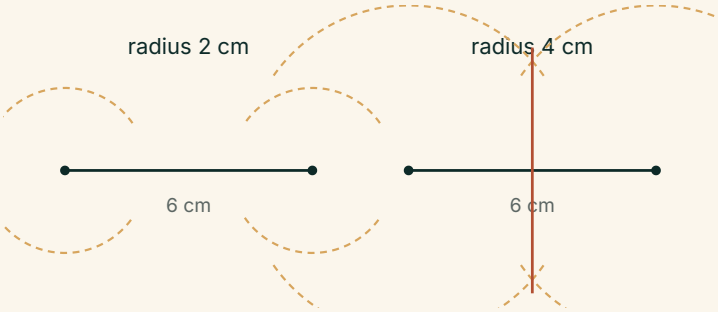
7.4.06
APPLICATION

A path must stay equally distant from gates A and B. Which construction locates the path? Explain how a partner could check one point on the path.



7.4.07
REASONING

A student uses a 2 cm radius to bisect a 6 cm segment. Compare the two diagrams. Explain why the first fails and the second works, and state the general radius condition.



7.4.08
REASONING

A construction looks perpendicular but the arcs were erased. Can you rely on appearance? Explain what evidence you would ask for and how you would check the result.

CHAPTER 7.4 · PRACTICE

Constructing triangles

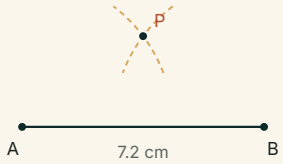
Practice 7.4 · Constructing triangles

7.4.09–7.4.10

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

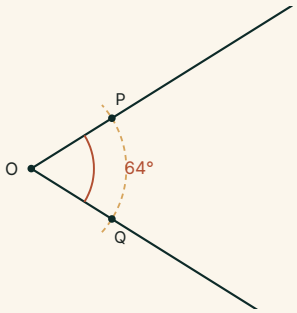
7.4.09
FLUENCY

Construct AB = 7.2 cm and its perpendicular bisector. Measure each half and state the check.



7.4.10
FLUENCY

Construct a 64° angle and bisect it with ruler and compasses. Measure both halves and state what they should be.



CHAPTER 7.4 · PRACTICE

Constructing triangles

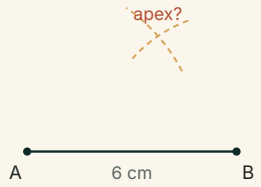
Practice 7.4 · Constructing triangles

7.4.11–7.4.12

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

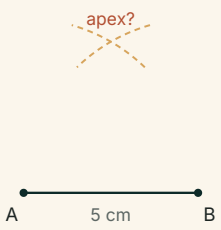
7.4.11 FLUENCY

Construct a triangle with sides 4 cm, 5 cm and 6 cm. Keep the arcs visible and label the vertices.



7.4.12 FLUENCY

Construct an equilateral triangle of side 5 cm. Explain why every angle must be 60°.



CHAPTER 7.4 · PRACTICE

Constructing triangles

Practice 7.4 · Constructing triangles

7.4.13–7.4.14

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

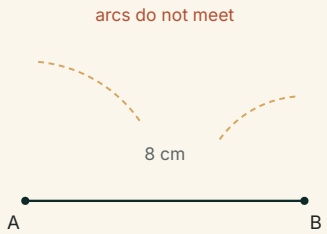
7.4.13 CHALLENGE

Construct an isosceles triangle with equal sides 6 cm and base 8 cm. Then construct the perpendicular bisector of the base. Explain why it passes through the apex.



7.4.14 REASONING

Attempt to construct a triangle with sides 3 cm, 4 cm and 8 cm. Describe what the arcs do and connect this to the triangle inequality.



CHAPTER 7.4 · PRACTICE

Constructing triangles

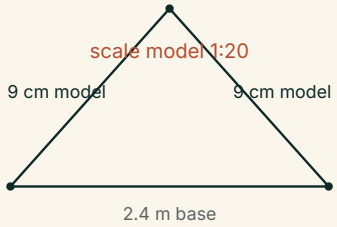
Practice 7.4 · Constructing triangles

7.4.15–7.4.16

Show a method, a representation and a check. Where a question asks for a reason, the reason is part of the answer.

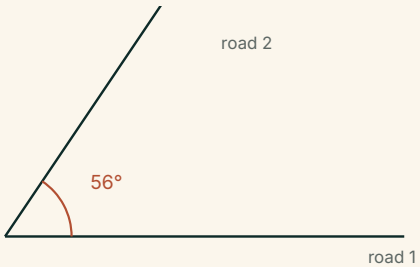
7.4.15
APPLICATION

A scale drawing uses 1 : 20. The real base of a triangular frame is 2.4 m. Find the model base in centimetres, then construct an isosceles model with equal sides 9 cm. What real rafter length does this represent?



7.4.16
CHALLENGE

A sprinkler is inside the angle between two straight roads that meet at 56° and must be equally distant from the roads. Sketch the locus and find the angle between the locus and each road. How would the answer change if the requirement were equal distance from two points instead?



CHAPTER 7 · MIXED PRACTICE

Connect ideas across the chapter

This route revisits items from the bank in a mixed order. Do not use a calculator. Draw a sketch, mark known facts and give a reason for each step.

ROUTE	ITEMS	FOCUS
Fluency	7.1.04, 7.1.05, 7.2.08, 7.3.06	Angle facts and regular polygons
Visual reasoning	7.2.03, 7.2.15, 7.3.05, 7.2.10	Diagrams, triangles and trusses
Construction	7.4.03, 7.4.06, 7.4.07, 7.4.08	Equal radii and evidence
Inquiry and challenge	7.3.13, 7.3.14, 7.3.15, 7.3.16	Triangle inequality and design

CHECK YOUR PROGRESS

Match the fact to the position: a line gives 180°, a point gives 360°, and co-interior angles are supplementary, not equal. Use the correct pair: alternate and corresponding angles are equal only when the crossed lines are parallel.

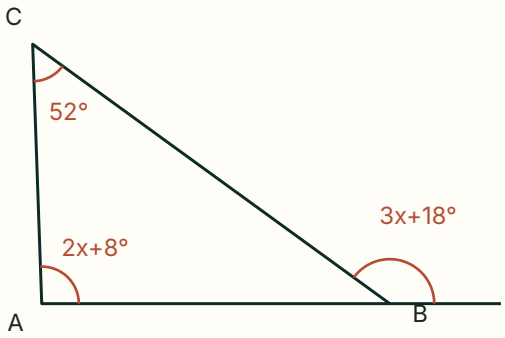
CHAPTER 7 · ASSESSMENT · CRITERION A

Knowing and understanding

Select appropriate mathematics, apply it successfully and solve problems in familiar and unfamiliar situations. Show your method and a check.

A1 · The badge design

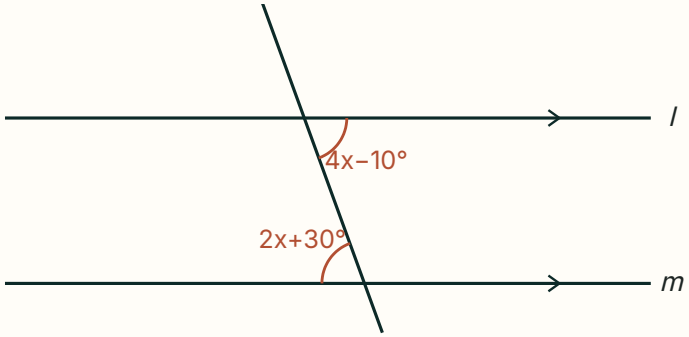
A badge is made from a triangle with one side extended. The exterior angle is $3x + 18°$, and the two opposite interior angles are $2x + 8°$ and $52°$. Find x, all three interior angles, and classify the triangle.



Assessment figure A1

A2 · Parallel supports

Two parallel supports are crossed by a brace. Alternate angles are labelled $4x - 10°$ and $2x + 30°$. Find x and the co-interior partner of one marked angle.



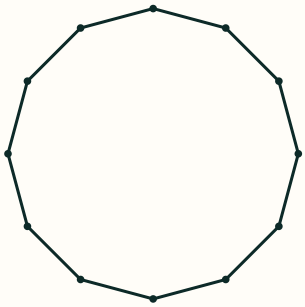
Assessment figure A2

CHAPTER 7 · ASSESSMENT · CRITERION A

Knowing and understanding · continued

A3 · A tile and a frame

A regular tile has interior angle 150°. Identify the polygon. A triangular frame has two sides 5 cm and 9 cm. Find all whole-centimetre lengths for the third side. Explain why the endpoints are excluded.



Assessment figure A3

Criterion A evidence: your choice of method, successful application and correct solutions. Do not just state answers; show the mathematics that supports them.

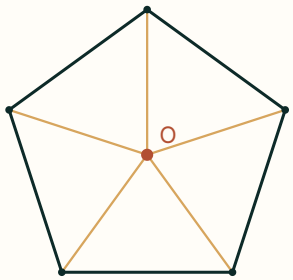
CHAPTER 7 · ASSESSMENT · CRITERION B

Investigating patterns

Work independently. Choose your own method and representation. Your task is to discover a relationship, describe it as a general rule, and verify it on new examples.

B1 · Triangles from an interior point

Choose a convex polygon. Mark a point O inside it and join O to every vertex. Investigate how the number of sides affects the interior angle sum.



One possible starting diagram

1. Investigate at least a triangle, quadrilateral, pentagon and hexagon. Organise your findings in a way of your choice.
2. Describe a general rule for an n-sided polygon using this interior-point method.
3. Verify your rule for an octagon using a different method.
4. Explain why the angles around O must be subtracted. What changes if O lies on a side instead of inside?

ATL EVIDENCE. Choose a technique, look for a pattern, generalise, verify and explain.

CHAPTER 7 · ASSESSMENT · CRITERION B

Workspace for investigation

Criterion B evidence: organised cases, a rule consistent with your findings, verification on a new example, and an explanation. A table is optional; choose a representation that makes the pattern visible.

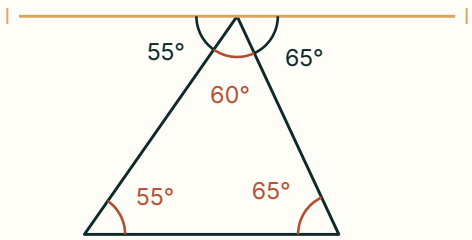
CHAPTER 7 · ASSESSMENT · CRITERION C

Communicating

Write for a partner who knows angle facts but has not seen this argument. Use mathematical language, a labelled diagram and a logical sequence.

C1 · Explain the triangle angle sum

Use the diagram to explain why the angles of any triangle sum to 180°. Your explanation must include the parallel line, alternate angles and the straight line through the apex.



Assessment figure C1

C2 · Improve a flawed explanation

A student writes: “The angles are 55°, 65° and 60°, so they add to 180°. Therefore every triangle adds to 180°.” Explain why this is not a general argument, then rewrite it using the parallel-line model.

A strong response uses correct notation, names the relationships, and moves logically from the diagram to the conclusion.

CHAPTER 7 · ASSESSMENT · CRITERION C

Workspace for communication

Self-check before submitting

- Have I used terms such as alternate, parallel and supplementary correctly?
- Does my diagram support each step?
- Could a partner follow the reasoning without seeing my working?

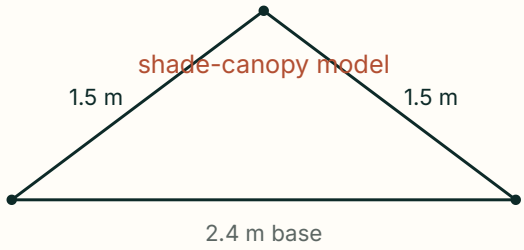
CHAPTER 7 · ASSESSMENT · CRITERION D

Applying mathematics in real-life contexts

Design brief: a simplified shade-canopy frame for a school courtyard. The dimensions below are illustrative, not construction specifications.

D1 · Check and draw the canopy frame

The frame is an isosceles triangle with base 2.4 m and equal rafters 1.5 m. A scale drawing uses 1 : 20.



Assessment figure D1 · simplified geometry only

1. Identify the relevant mathematical elements of the brief.
2. Select and apply strategies to check whether the triangle is possible and to find the scale-drawing lengths.
3. Construct the scale triangle with ruler and compasses. Measure the apex angle to the nearest degree and check the angle sum.
4. Explain the degree of accuracy of your drawing and measurements.
5. Describe whether the solution makes sense in context, and state one limitation of the model.

CHAPTER 7 · ASSESSMENT · CRITERION D

Workspace for application and interpretation

Context check

This task assesses geometry, scale and interpretation. It does not certify a real structure. Real canopies also need material thickness, joints, loads, wind and safety checks.

Criterion D evidence: relevant elements, appropriate strategies, a successful solution, accuracy, and interpretation in context.

CHAPTER 7 · REVIEW

Summary and connections

Angle facts

- Straight line: 180°.
- Full turn: 360°.
- Vertically opposite: equal.
- Parallel lines: corresponding and alternate equal; co-interior supplementary.

Triangles and polygons

- Triangle sum: 180°.
- Exterior angle: sum of opposite interior angles.
- n-gon interior sum: $(n-2)\times180°$.
- Exterior sum: 360°.

Constructions

- Perpendicular bisector: equal radii greater than half the segment.
- Angle bisector: equal arcs from the arms.
- SSS triangle: arcs meet when the triangle inequality holds.

Inquiry

- Triangle inequality: two shorter sides > longest.
- Equality gives a degenerate straight line.
- Keep arcs visible as evidence.

CONNECTIONS · DRAWING IS NOT PROVING

A construction is an argument made with instruments. Each arc records a fixed distance from a named point, so the finished drawing already contains evidence: equal radii, equal distances, and a property that can be checked. A picture that looks right is not enough; the defining relationship must hold.

Where this leads: Chapter 8 turns classification into a precise language. Sets describe which objects share a property, and Venn diagrams show relationships of overlap and exclusion. Chapter 9 then measures the shapes whose boundaries you have learned to reason about.

CHAPTER 7 · CHECK YOUR PROGRESS

Reflection and next step

SUCCESS CRITERION	MY RATING (1–4)	EVIDENCE: ITEM OR TASK	NEXT STEP
I can name and classify angles.			
I can use angle facts with reasons.			
I can calculate triangle and polygon angles.			
I can carry out and explain constructions.			
I can decide whether three lengths form a triangle.			
I can produce and evaluate a scale drawing.			

REFLECTION

Which inquiry question changed your thinking most: the factual question about angle totals, the conceptual question about why the inequality must hold, or the debatable question about when a construction is reliable enough to build from? Use one item from the bank as evidence.

ATL EVIDENCE. In order for a student to think creatively and critically, students must choose an appropriate geometric fact or construction, test a design and explain why the method is reliable. Point to the place in your work where you did this.

Review the answer key only after attempting each question. A correct answer without a reason is only half an explanation.

CHAPTER 7 · ANSWERS AND REASONING

Answers 7.1.01–7.1.05

7.1.01 **35° acute; 90° right; 125° obtuse; 180° straight; 210° reflex.**
Acute is between 0° and 90°; obtuse between 90° and 180°; reflex between 180° and 360°. The exact values 90° and 180° have their own names.

7.1.02 **The vertex is B; the rays are BA and BC. Sketches may vary but the angle at B must be obtuse.**
The middle letter names the vertex. Many correct sketches are possible.

7.1.03 **Straight line: 180°; full turn: 360°; vertically opposite angles: equal.**
A straight line is a half-turn; a point is a full turn; crossing lines make equal opposite pairs.

7.1.04 **x = 115°.**
180° − 65° = 115°; check 65° + 115° = 180°.

7.1.05 **85°.**
360° − (90° + 110° + 75°) = 85°; the four angles sum to 360°.

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Answers 7.1.06–7.1.10

7.1.06 **x = 65°, y = 115°, z = 65°.**
y is vertically opposite 115°. x and z are adjacent to 115° on straight lines, so each is 180° − 115°.

7.1.07 **x = 35°; the angles are 65°, 70° and 45°.**
(x+30)+2x+(x+10)=180, so 4x+40=180 and x=35. Check 65+70+45=180.

7.1.08 **x = 15°; each marked angle is 75°; an adjacent angle is 105°.**
4x+15=2x+45 gives 2x=30, x=15. Adjacent angles on a line sum to 180°.

7.1.09 **78°.**
Vertical is perpendicular to level ground, so the ground–post angle is 90°. The ramp splits this right angle into 12° and x, so x = 90° − 12° = 78°.

7.1.10 **60°; no, the order does not change the total.**
360° − (118° + 97° + 85°) = 60°. Addition is commutative, so the missing part of a full turn is unchanged.

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Answers 7.1.11–7.1.15

7.1.11 **36°, 72°, 108°, 144°; no pair is complementary.**
The ratio has 10 parts, so one part is 360°÷10=36°. Complementary angles sum to 90°; no two of these angles do.

7.1.12 **Claim 2 is wrong: angles around a point sum to 360°, not 180°. For example 90°, 90°, 100° and 80° meet at a point.**
A half-turn is 180°; a full turn is 360°. Many counterexamples are possible.

7.1.13 **True.**
If each angle is x, then 2x=180°, so x=90°. The sketch should show two right angles on a line.

7.1.14 **2x + (x+40) + 80 = 360; 3x = 240; x = 80°. Angles are 160°, 120° and 80°.**
Collect like terms: 3x+120=360. Check 160+120+80=360.

7.1.15 **x = 50°; angles 50°, 70°, 100°, 140°; largest is 140°, which is obtuse because it is between 90° and 180°.**
x+(x+20)+2x+(3x−10)=360 gives 7x+10=360.

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Answers 7.1.16–7.2.04

7.1.16 Every correct diagram has angles 70°, 110°, 70°, 110°. Orientation and the lengths of the drawn segments may change.
Vertically opposite angles are equal and adjacent angles are supplementary. The angle family is fixed even when the drawing is rotated.

7.2.01 Examples: corresponding angle 1 and angle 5; alternate angle 4 and angle 5; co-interior angle 4 and angle 6. Corresponding and alternate pairs are equal; co-interior pairs are supplementary.
Other correct pairs include angle 2/angle 6, angle 3/angle 6 and angle 3/angle 5. The parallel marks are essential.

7.2.02 False. For example angle 4 and angle 6 are co-interior and supplementary, not necessarily equal.
Only corresponding and alternate pairs are equal. A counterexample is any non-right co-interior pair.

7.2.03 a = 68°, b = 112°, c = 68°.
a is alternate to 68°; b is co-interior with 68°; c corresponds to 68° (angle 8).

7.2.04 angle 5 = 52°, angle 6 = 128°, angle 7 = 128°.
angle 1 is adjacent to angle 2 on a line, so angle 1=52°; angle 5 corresponds to angle 1. angle 6 corresponds to angle 2; angle 7 is vertically opposite angle 6.

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Answers 7.2.05–7.2.09

7.2.05 Yes: 112° + 68° = 180°, so the lines are parallel by the converse of the co-interior angle fact.
Equal alternate angles or supplementary co-interior angles can establish parallel lines. A sketch without those facts does not prove parallelism.

7.2.06 60°; acute scalene.
180°−35°−85°=60°. All angles are acute and different, so the sides are different.

7.2.07 x = 30°; angles 30°, 60°, 90°; right triangle.
x+2x+3x=180°, so 6x=180°.

7.2.08 70° each.
Equal side ticks show an isosceles triangle. (180°−40°)÷2=70°.

7.2.09 67° and 65°.
Other opposite angle =115°−48°=67°. Adjacent interior angle =180°−115°=65°.

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Answers 7.2.10–7.2.14

7.2.10 Apex 50°; exterior 115°; isosceles, so the rafters are equal.
180°−65°−65°=50°; 180°−65°=115°. Equal base angles imply equal opposite sides.

7.2.11 x = 114°.
Co-interior angles between parallel lines sum to 180°: 180°−66°.

7.2.12 Marked base angle 70°; whole left-tip angle 140°.
Each base angle is (180°−40°)÷2=70°. The lower triangle is a mirror copy of the upper triangle, so it has the same base angle; the whole tip is 70°+70°.

7.2.13 Top interior angle 76°; exterior angle 104°.
180°−90°−14°=76°. The exterior angle is supplementary to 76°, so 104°.

7.2.14 No.
Two obtuse angles already sum to more than 180°, leaving no positive third angle. A sketch will fail to close as a triangle.

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Answers 7.2.15–7.2.19

7.2.15 The exterior angle equals the sum of the opposite interior angles: 110°=60°+50°. The adjacent interior angle is 70°.

The exterior angle and the adjacent interior angle are supplementary. The two opposite interior angles are not added to the exterior angle to make 180°.

7.2.16 The angle sets are the same (40°, 60°, 80°); the sizes are different. Equal angles do not force equal side lengths.

Angle sum controls the angles, not the overall scale. The diagrams are a counterexample to “same angles means same side lengths”.

7.2.17 Base angles 50° each; apex 80°.

2x+(x+30)=180 gives x=50. Checks: 50+50+80=180; apex exceeds a base angle by 30°.

7.2.18 52°, 78° and 50°; acute scalene.

Opposite interior angles sum to 130°. Five ratio parts give 26° each: 52° and 78°. Adjacent interior angle is 180° −130°=50°.

7.2.19 Angles 1, 4, 5 and 8 are 73°; angles 2, 3, 6 and 7 are 107°. Four angles are acute and four are obtuse.

At each intersection the acute angles are vertically opposite and the obtuse angles are vertically opposite; adjacent angles sum to 180°. Parallel-line facts transfer the values to the second intersection.

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Answers 7.2.20–7.3.04

7.2.20 x = 25°; marked angles 60°; co-interior partner 120°.
Alternate angles are equal: 2x+10=3x−15, so x=25. Co-interior angles sum to 180°.

7.3.01 Pentagon 5; hexagon 6; octagon 8. A regular polygon has all sides equal and all interior angles equal.
Polygon names count sides. Both equal sides and equal angles are required for regularity.

7.3.02 Both have six sides and interior angle sum 720°. A regular hexagon has equal sides and equal 120° angles; an irregular hexagon need not.
The angle-sum rule depends on the number of sides, not regularity. Sketches may vary.

7.3.03 Quadrilateral: 2 triangles, 360°; pentagon: 3, 540°; hexagon: 4, 720°.
An n-gon splits into n−2 triangles from one vertex; multiply by 180°.

7.3.04 Heptagon 900°; nonagon 1260°; decagon 1440°.
(7−2)×180=900; (9−2)×180=1260; (10−2)×180=1440.

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Answers 7.3.05–7.3.09

7.3.05 x = 80°.
360°−(110°+75°+95°)=80°.

7.3.06 8 sides; interior 135°.
360°÷45°=8; interior and exterior are supplementary, so 180°−45°=135°.

7.3.07 Regular octagon. No: 360°÷135° is not a whole number, so whole tiles cannot fill a full turn exactly.
Exterior angle is 45°, giving 8 sides. Two tiles leave 90°; three overlap. This is an angle argument, not a claim about every possible tiling with other shapes.

7.3.08 Total turn is 360° in each row because the walker returns to the starting direction after one full circuit.
The number of turns changes, but one complete circuit is always one full turn.

7.3.09 3,4,8: no (7<8); 5,5,9: yes (10>9); 6,7,10: yes (13>10).
A triangle needs the two shorter sides together to exceed the longest. Equality would give a straight line.

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Answers 7.3.10–7.3.14

7.3.10 Rows: yes, no, no (borderline), yes. Student row may vary; a borderline row has shorter sum equal to longest.
Compare a+b with c when c is longest. Strictly greater gives a triangle; equal gives zero area; less gives a gap.

7.3.11 Yes, it already is a triangle because 7+9>15. No change is needed.
The question checks whether you test before changing. The shortest-sum test is 16>15.

7.3.12 Impossible: the two shorter sides give 4+4<9. The arcs do not meet; there is a 1 cm gap.
The relevant test compares the two shorter sides with the longest. A correct sketch shows the failure to close.

7.3.13 5+8=13 gives a degenerate straight line. With 12 cm, 5+8>12, so a triangle is possible.
Equality places the apex on the base and gives zero area. Reducing the base lets the arcs meet above it.

7.3.14 5, 6, 7, …, 17 cm.
The third side x must satisfy 11−7 < x < 11+7, so 4<x<18. Endpoints are excluded.

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Answers 7.3.15–7.4.02

7.3.15 5 m.
2x>8, so x>4. At x=4 the rafters lie flat along the base.

7.3.16 If the base is c, arcs of radii a and b meet above the base only when a+b>c. Equality meets on the base; less leaves a gap.
The arc intersection is the possible apex. No intersection above the base means no non-degenerate triangle.

7.3.17 12 sides.
Interior + exterior =180°. If interior=5e, then 6e=180°, e=30°, n=360°÷30°=12.

7.4.01 Perpendicular bisector for two points; angle bisector for the angle between two roads.
A perpendicular bisector compares distances to endpoints. An angle bisector compares perpendicular distances to the two arms.

7.4.02 They record equal distances from the named centres. Changing the radius destroys the equal-distance evidence.
The construction is justified by the defining property, not by appearance.

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Answers 7.4.03–7.4.07

7.4.03 ii, iv, i, iii; radius > 4 cm.
The arcs must meet at two points. The line through those points is perpendicular to AB and passes through its midpoint.

7.4.04 34° each.
The same radius from the two arm-intersection points locates a point on the bisector; the two half-angles are equal.

7.4.05 x = 12°.
Each half is 42°, so 3x+6=42 and x=12. Check 3(12)+6=42.

7.4.06 The perpendicular bisector of AB. Check that the point is the same distance from A and B.
Every point on the perpendicular bisector is equidistant from the endpoints.

7.4.07 Radius must be greater than 3 cm. A 2 cm radius is too small; 4 cm works.
Equal-radius arcs meet at two points only when the radius exceeds half the segment length.

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Answers 7.4.08–7.4.12

7.4.08 No. Ask for the equal-radius arcs or re-construct them; check midpoint, right angle and equal distances to A and B.
A drawing is not proof. The defining properties and construction evidence justify the result.

7.4.09 Each half is 3.6 cm; the bisector meets AB at 90°.
Use one radius greater than 3.6 cm from both endpoints. Keep arcs visible.

7.4.10 32° each.
Draw an arc from the vertex, then equal-radius arcs from the two arm intersections. Join the vertex to their intersection.

7.4.11 A correct SSS construction; for example base 6 cm, arcs 4 cm and 5 cm from the endpoints.
Check the triangle inequality first: 4+5>6. The arc intersection fixes the third vertex.

7.4.12 All sides 5 cm; all angles 60°.
Equal sides give equal opposite angles; 180°÷3=60°.

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Answers 7.4.13–7.4.16

7.4.13 The apex is equidistant from the base endpoints, so it lies on the perpendicular bisector.
Use radius 6 cm from both base endpoints. The perpendicular bisector is the locus of points equidistant from the endpoints.

7.4.14 The arcs do not meet because 3+4<8.
No apex exists; the two shorter sides cannot reach across the base.

7.4.15 Model base 12 cm; real rafters 1.8 m.
2.4 m=240 cm; 240÷20=12 cm. Model sides 9 cm represent 9×20=180 cm=1.8 m.

7.4.16 The angle bisector; 28° to each road. For two points, use the perpendicular bisector of the segment joining them.
Distance to lines is measured perpendicular to the lines; the angle bisector is the locus. Distance to points uses the perpendicular bisector.

CHAPTER 7 · ASSESSMENT GUIDANCE

Sample responses · Criteria A and B

A1 x = 42°; interior angles 92°, 36°, 52°; obtuse scalene.
3x+18 = (2x+8)+52 gives x=42. The adjacent interior angle is 180−144=36°.

A2 x = 20°; marked angles 70°; co-interior partner 110°.
4x−10 = 2x+30 gives x=20.

A3 Regular dodecagon; third side 5–13 cm.
Exterior angle 30°, so n=12. For sides 5 and 9, 4<x<14; endpoints give degenerate cases.

B1 Interior-point method: n triangles, total 180n°, subtract the 360° around O, giving 180n−360 = (n−2)180°.
For an octagon this gives 1080°, matching (8−2)×180°. If O lies on a side, the angles around O on the interior side sum to 180°, and the count of triangles changes; the same interior sum must still result.

These are sample responses, not the only acceptable methods. Criterion B credit depends on the student's own investigation, generalisation and verification.

CHAPTER 7 · ASSESSMENT GUIDANCE

Sample responses · Criteria C and D

C1–C2 A general argument uses the parallel line through the apex.

Alternate angles transfer the two base angles to the apex. The three angles on the straight line through the apex sum to 180°, so the triangle angles sum to 180°. One numeric example alone is not a general argument.

D1 Possible: 1.5+1.5>2.4. Scale lengths: base 12 cm, rafters 7.5 cm.

2.4 m=240 cm; 240÷20=12 cm. 1.5 m=150 cm; 150÷20=7.5 cm. A measured apex is about 106° and base angles about 37°; accept small measurement variation. Accuracy: nearest millimetre in the model corresponds to 2 cm in reality; angles to nearest degree. Limitations include thickness, joints, loads and wind.

Criterion C values clear language, representations and logical organisation. Criterion D values context, strategy, accuracy and interpretation, not just a correct calculation.